CLONED_KMKK

PART A:

1.

a. $a = \frac{1}{2}$, f is not continuous at $x = 3$

b. $\frac{1}{6}$

2.

a. 2x + 2

b. $$\frac{d^2y}{dx^2} = \left(\frac{3}{2}\right)x^{-\frac{5}{2}} + \left(\frac{1}{4}\right)x^{-\frac{3}{2}}$$
$$\frac{d^3y}{dx^3} = -\left(\frac{15}{4}\right)x^{-\frac{7}{2}} - \left(\frac{3}{8}\right)x^{-\frac{5}{2}}$$

c. $$-\sin(x+y)\left(1+\frac{dy}{dx}\right) + 2\sin y + 2x\cos y\frac{dy}{dx}$$

3. $\left(\frac{\pi}{2}, -1\right)$ is a relative minimum point.

PART B

1. x= 1, y = -2, -1.107

2.

a. x = ln 4

b. $[-2,-1) \cup (3,6]$

3.

a. Since $x_1 \neq x_2$, $f(x)$ is not a one-to-one function. Therefore, the function f does not have an inverse.

b. M = 2

4.

a.

i. Shown

$\therefore g^{-1}(x) = e^{2x} + 1$

ii. $\therefore f(x)$ and $g(x)$ are inverse to each other's.

iii. $h(x) = 2x^2$

b.

i. $3\sin(x-\pi)$

ii. Graph

iii. $R_{f \circ g} = [-3, 3]$

5.

a.

i. 0

ii. -2

b. m = -2, n = 8

6.

a. f is not differentiable at x = 1

b. $\frac{dy}{dx} = \frac{y}{x(2y^2-6)}$

c.

i. $-\frac{(4t+1)^2}{(2t+1)}$

ii. $t = -\frac{3}{8}$

7. $\frac{1}{40} cms^{-1}$